Development of a Wearable Speech-to-Text Emotion and Speaker Recognition System for Individuals with Hearing Impairments

1. Prototype Description

1.1 Overview

This project presents a wearable assistive-technology prototype for individuals with hearing impairments. The system pairs a Raspberry Pi 4 (RPi 4) with an ESP32-based wrist-worn device to recognise vocal emotion and speaker profile in real time. The wearable shows the detected emotion and speaker label on an OLED display and plays a distinct vibration pattern for each emotion. By translating vocal emotion into visual and tactile signals, the device supports emotional awareness and social interaction for its users.

This prototype is the hardware extension of the system published in the *Proceedings of the 6th International Conference on Electronics and Sustainable Communication Systems (ICESC-2025), IEEE*, pp. 1056–1064.

1.2 System Architecture

The system has two main parts: a Raspberry Pi 4 that handles audio processing and recognition, and an ESP32-based wearable that delivers feedback to the user.

1.2.1 Raspberry Pi 4: Audio Processing and Recognition Unit

- **Microphone input:** A microphone connected to the Raspberry Pi 4 continuously records speech and ambient audio.
- **Speaker recognition:** A 1D Convolutional Neural Network (CNN) classifies MFCC features of the audio into one of four speaker profiles (A1–A4).
- **Emotion recognition:** A pretrained Wav2Vec2 model classifies the speaker's emotional state as one of five states: **Neutral**, **Happy**, **Sad**, **Angry** or **Fearful**.
- **Wireless communication:** The results are sent to the wearable over Wi-Fi using HTTP or MQTT.

1.2.2 ESP32-Based Wearable: Feedback Interface

The ESP32 module is built into a wrist-worn device that gives the user real-time feedback. Its key features are:

- **ESP32 microcontroller:** Receives results from the Raspberry Pi over Wi-Fi and controls the onboard hardware.
- **OLED display:** Shows the detected emotion and the speaker label (for example, "Person A") as readable text.
- **Haptic motor:** Plays a distinct vibration pattern for each emotion (Neutral, Happy, Sad, Angry, Fearful), so the user can *feel* the emotional tone of speech.

- **Battery and charging circuit:** A rechargeable battery with an onboard charging circuit allows extended, portable use.
- **Compact design:** Comparable in size to a smartwatch or fitness tracker, for comfort and ease of daily use.

1.3 System Workflow

1 The microphone on the Raspberry Pi 4 captures live speech.

2 MFCC features are extracted and the CNN identifies the speaker profile (A1–A4).

3 The Wav2Vec2 model classifies the emotion as Neutral, Happy, Sad, Angry or Fearful.

4 The speaker label and emotion are sent over Wi-Fi to the ESP32 wearable.

5 The wearable displays the emotion and speaker label on the OLED screen and plays the vibration pattern assigned to that emotion.

Together, these steps address a shared difficulty for people with hearing impairments: not being able to sense the emotional tone of a conversation. By translating vocal cues into visual and tactile form, the device gives users an immediate, wordless sense of how something is being said and who is saying it.

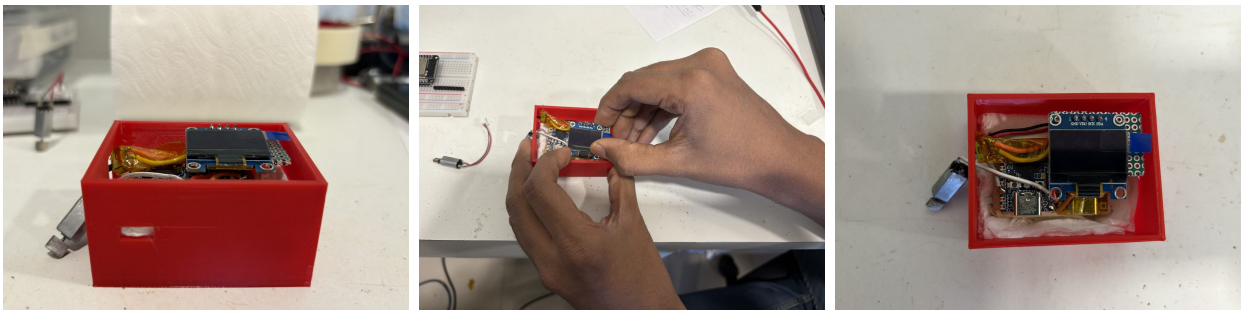


Fig. 1. *The wearable prototype at successive stages of assembly.*

1.4 Speaker Recognition and Emotion Feedback

The system differentiates speakers to make conversations more accessible for people with hearing impairments. A Convolutional Neural Network (CNN) processes Mel-frequency cepstral coefficients (MFCCs) extracted from the audio and classifies each voice into one of four speaker profiles: A1 (older male), A2 (older female), A3 (younger male) and A4 (younger female), displayed on the wearable as Person A, B, C and D. In scenario-based evaluations with different voices and speaking styles, the CNN achieved **99.3% accuracy** in distinguishing these speaker profiles.

Once a speaker is identified, the corresponding label (for example, "Person A") appears on the OLED display of the ESP32 wearable. This helps the user tell speakers apart in multi-speaker settings such as group discussions and public spaces, even without knowing who the people are beforehand.

In parallel, a pretrained Wav2Vec2 model detects the speaker's emotion. The emotion is shown on the OLED display and, crucially, conveyed through the **haptic motor**: each emotion (Neutral, Happy, Sad, Angry, Fearful) triggers its own vibration pattern. The patterns are simple and easy to tell apart, so users can feel the emotional tone of speech even where the screen is hard to see, such as outdoors or in

noisy environments. Visual and haptic feedback together achieved an **average response time of 250 ms**, enabling smooth, real-time interaction.

2. Design Decisions

Consulting healthcare experts guided the design towards real user challenges and kept the system practical. The key decisions were to combine visual and tactile feedback and to prioritise a user-centred design:

- **Custom enclosure:** Designed a small 3D-printed enclosure for a more ergonomic fit than generic boxes.
- **Raspberry Pi 4:** Selected for its 4 GB RAM and processing power, which let the CNN and Wav2Vec2 models run on-device.
- **ESP32 integration:** Chosen for low-power wireless communication and control of the haptic motor and display.
- **Testing approach:** Conducted scenario-based tests with diverse audio samples to check robustness across voices and speaking styles.

3. Creative Thinking

I designed emotion-specific vibration patterns so that users could feel whether someone sounded neutral, happy, sad, angry or fearful, and paired them with an on-screen emotion label for quick recognition in noisy or low-light environments. These features came from putting myself in my users' position rather than from a tutorial.

To simulate different speakers, I built a dataset from audiobook clips and transformed them with Speechify AI into four voice profiles varying in age and gender, an idea that came from observing how voice tone changes in real life. Instead of building an emotion model from scratch, I integrated the pretrained Wav2Vec2 model with my CNN-based speaker recognition, keeping average feedback time to 250 ms. Balancing originality with efficiency in this way was a key design choice. Table 1 summarises these elements.

Creative element	Description	Why it is distinctive
Emotion-specific haptic feedback	Each emotion (Neutral, Happy, Sad, Angry, Fearful) triggers its own vibration pattern.	Turns emotional tone into a physical signal users can feel, not just read.
On-screen emotion and speaker labels	The OLED shows the detected emotion and the speaker label (Person A–D).	Adds emotional and speaker context to conversation at a glance.
Dataset generation with Speechify AI	Created four voice profiles (older/younger, male/female) from audiobook clips.	Simulated real-world voice diversity without costly recordings.
Wav2Vec2 + CNN integration	Pretrained Wav2Vec2 for emotion and a custom CNN for speaker profiles, run in parallel.	Reuses a state-of-the-art model to cut training time while keeping feedback real-time.
Compact wearable design	Wrist-worn device combining ESP32, OLED display and vibration motor.	Merges function and comfort into assistive technology suitable for daily wear.
Dual feedback (visual + tactile)	Immediate response through both the screen and haptic vibration.	Remains usable in loud or low-visibility settings.

Table 1. *Creative elements of the prototype.*

My curiosity pushed me to go from a breadboard to a compact wearable. From designing the vibration feedback circuit to coding responsive on-device models, I did not just build a prototype; I designed an experience grounded in practicality and empathy.

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Appendix A: Prototype Design and Build

The figures below document the prototype from breadboard testing to the finished wearable.

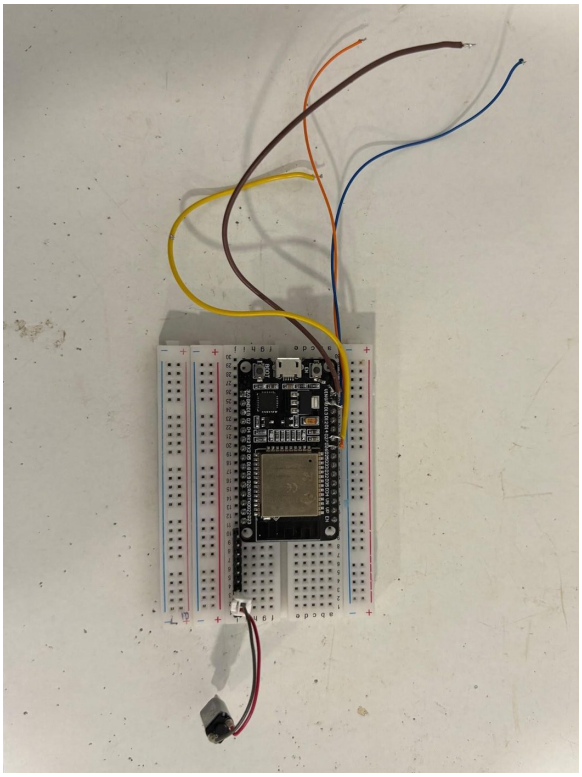


Fig. 2. *Designing the circuit on a breadboard: ESP32 with the vibration motor connected.*



Fig. 3. *Designing the prototype: fitting the OLED display into the enclosure.*

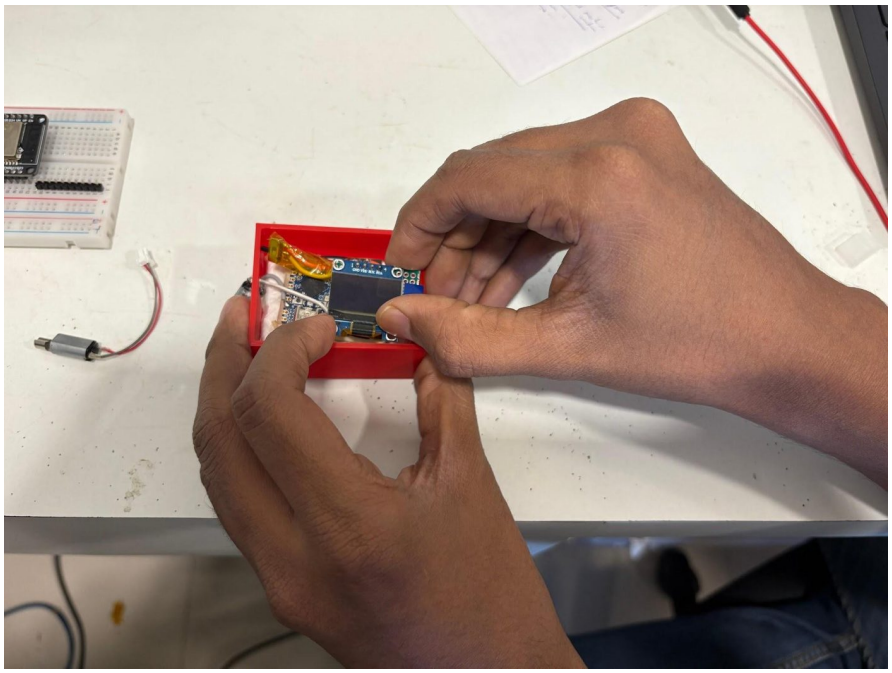


Fig. 4. *Assembling the prototype.*

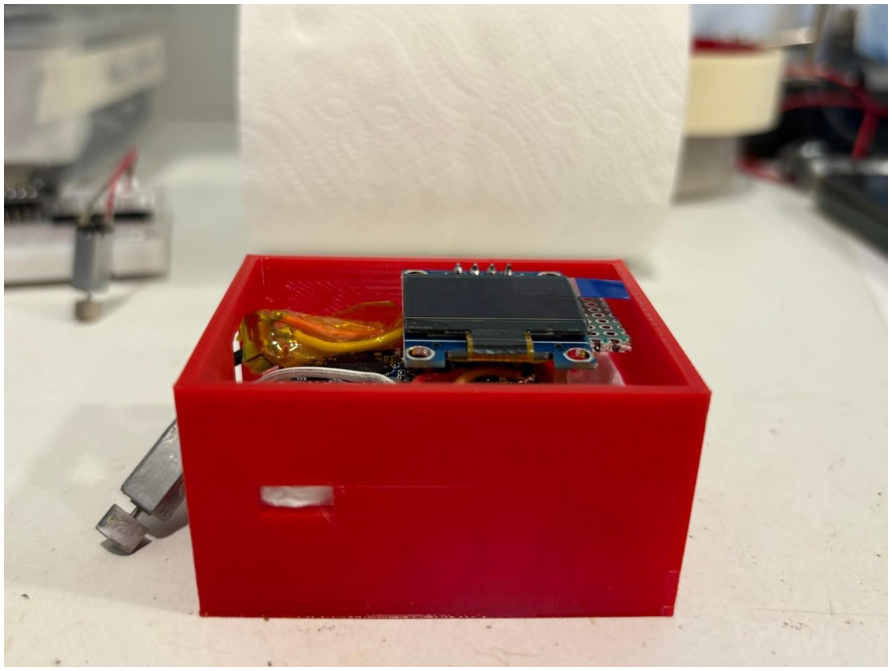


Fig. 5. *The completed prototype.*

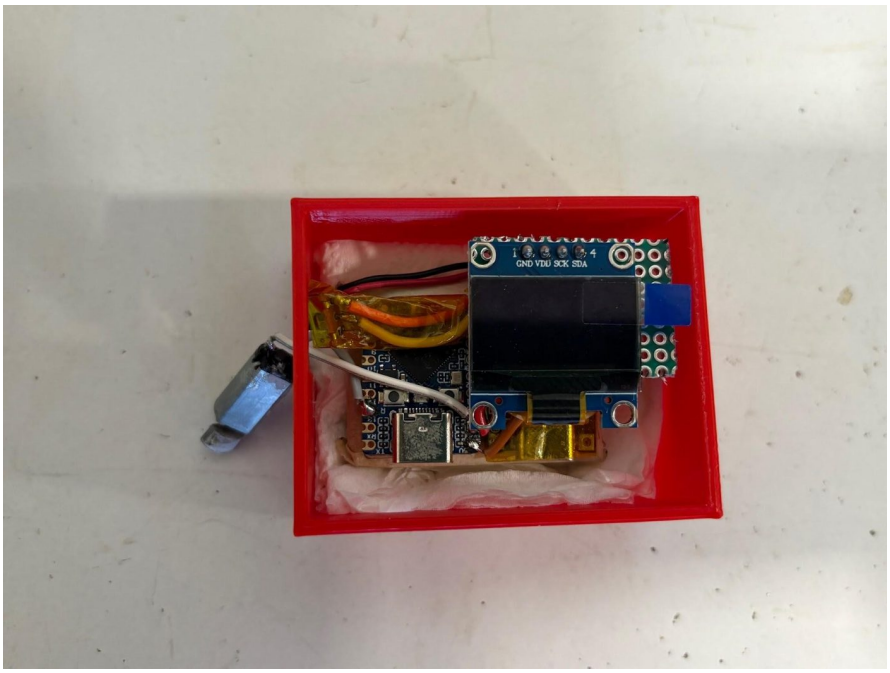


Fig. 6. *Components inside the prototype: ESP32, OLED display, battery and charging circuit.*

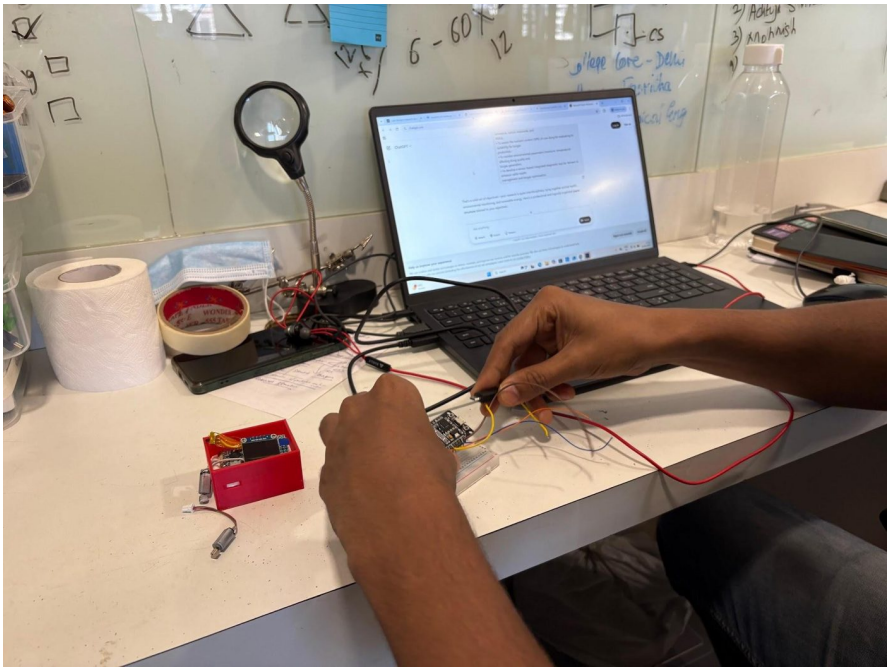


Fig. 7. *Testing the prototype.*